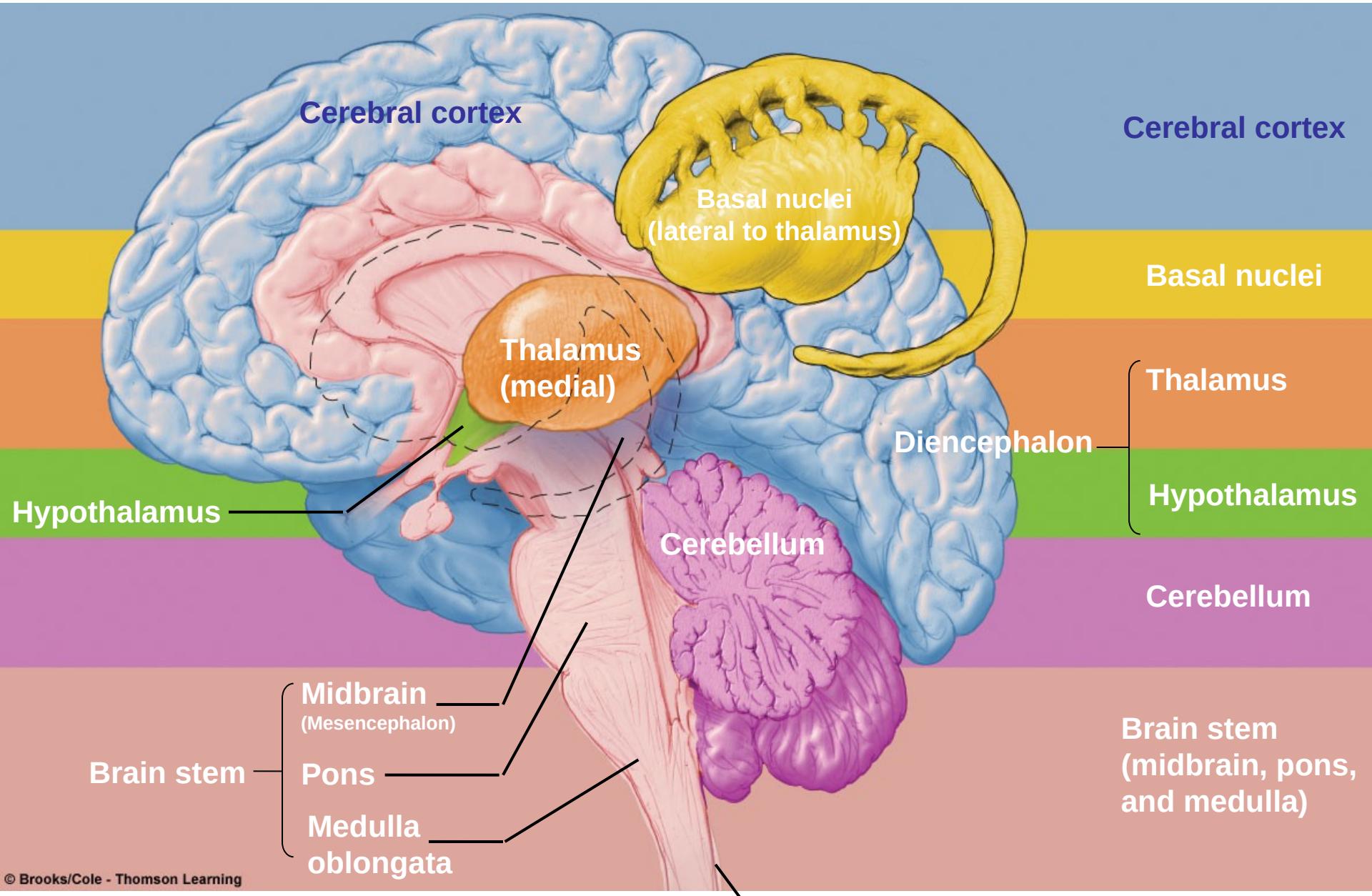
Cerebellum

LEARNING OBJECTIVES

At the end of this lecture, the students will be able to :

- Narrate the Gross anatomy of the cerebellum
- Understand the various terms like folia, vermis, tracts and nuclei of cerebellum
- Major efferent and afferent pathways and their function
- Human diseases associated with cerebellar dysfunction

Brain components



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Some Terminologies

- White matter : myelinated fibre tracts
- Gray matter : areas of neuronal cell bodies
- Tracts : collections of axons subserving similar function or location in CNS
- Nerves : peripheral axons
- Nucleus : collection of neurons subserving similar function in CNS – e.g., caudate nucleus, red nuclei
- Brainstem: Midbrain (Mesencephalon) + Pons + Medulla Oblongata
- Folia : Leaves
- Vermis: Worm

Cerebellum — *External Configurations*

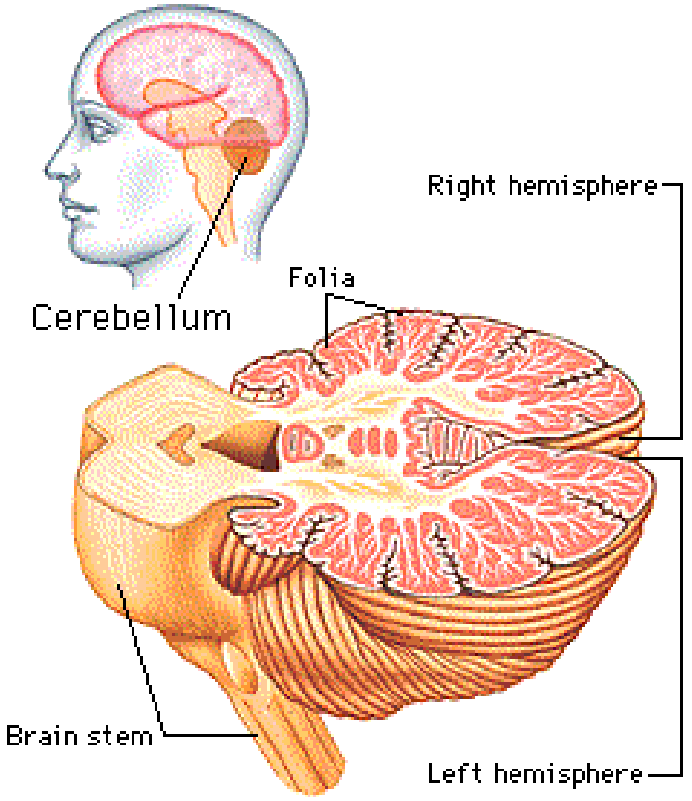
- **located in posterior cranial fossa**
- **tentorium cerebelli (cerebrum), 4th ventricle (brain stem)**
- **communicate with other structure via**
  superior, middle, and inferior cerebellar peduncle

- ***longitudinal division***
  Vermis, Paravermal Region, Cerebellar Hemisphere

- ***transverse division***
  Anterior Lobe
  ---------- *primary fissure*
  Posterior Lobe
  ---------- *posterolateral fissure*
  Flocculonodular Lobe

The Cerebellum

- Located dorsal to the pons and medulla
- Makes up 11% of the brain's mass
- Cerebellar activity occurs subconsciously
- Provides precise timing and appropriate patterns of skeletal muscle contraction
  Programming ballistic movements
- Acts as comparator for movements
- Correction of ongoing movements
- Motor learning
  Shift from conscious ---> unconscious



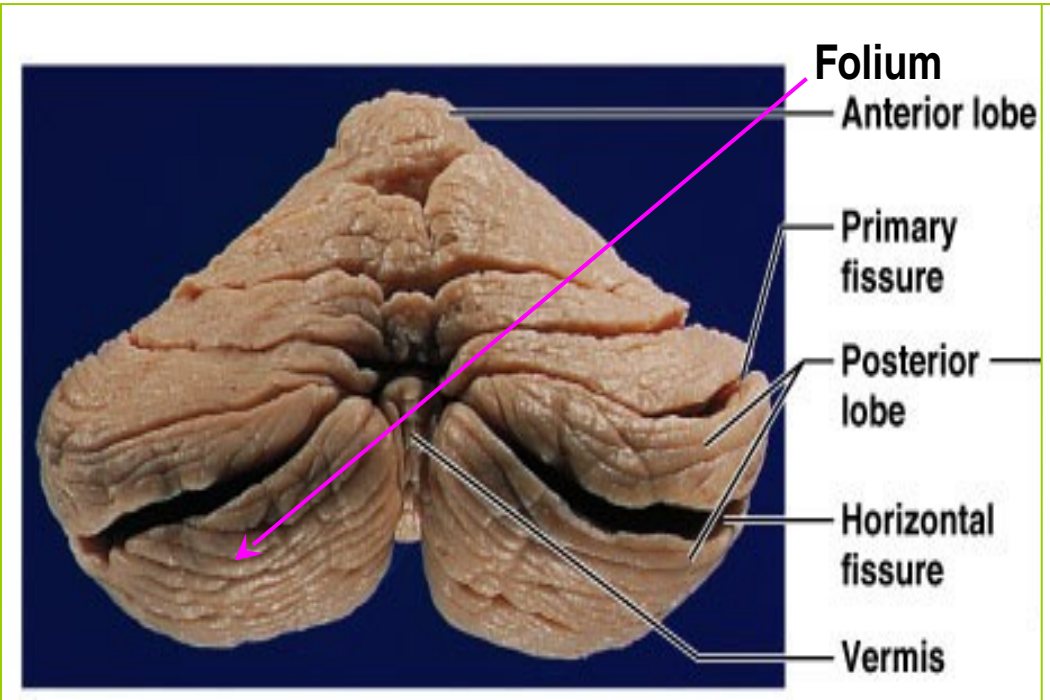
Anatomy of the Cerebellum

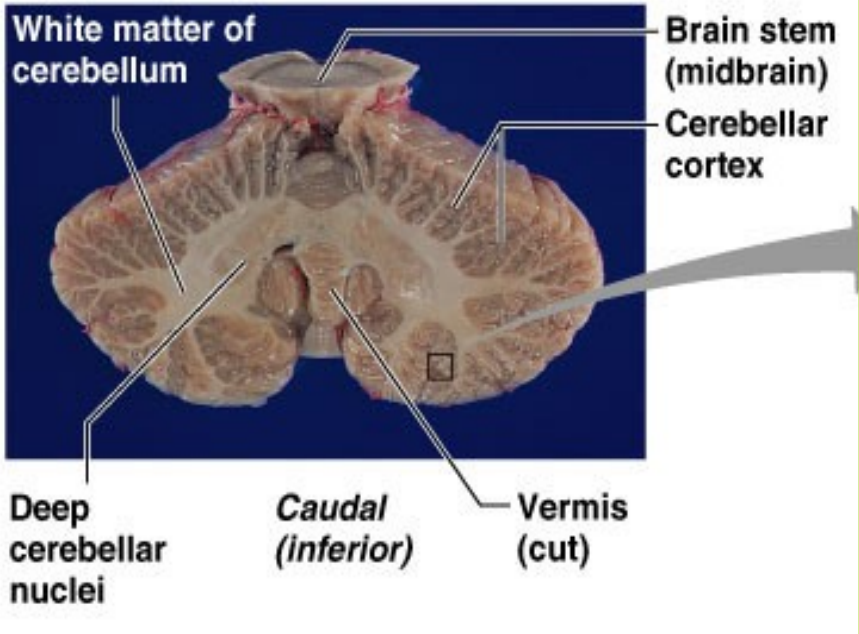
2 symmetrical hemispheres connected medially by the Vermis

Folia: Transversely oriented gyri

3 lobes in each hemisphere: Anterior, Middle , Flocculonodular (FN)

Anterior lobe found on superior surface and is seperated from the middle lobe by fissura prima





Anatomy of the Cerebellum

- The middle lobe is situated b/w the fissura prima and the uvulonodular fissure
- The flocculo nodular fissure is situated oposterior to the uvulonodular fissure
- A deep horizontal fissure seperates the superior from inferior surface
- Neural arrangement: Gray matter (Cortex), White matter (Internal),
  Scattered cerebellar nuclei: dentate, globose, emboliform, fastigial
- Arbor vitae (tree of life): distinctive treelike pattern of the white matter

Cerebellum

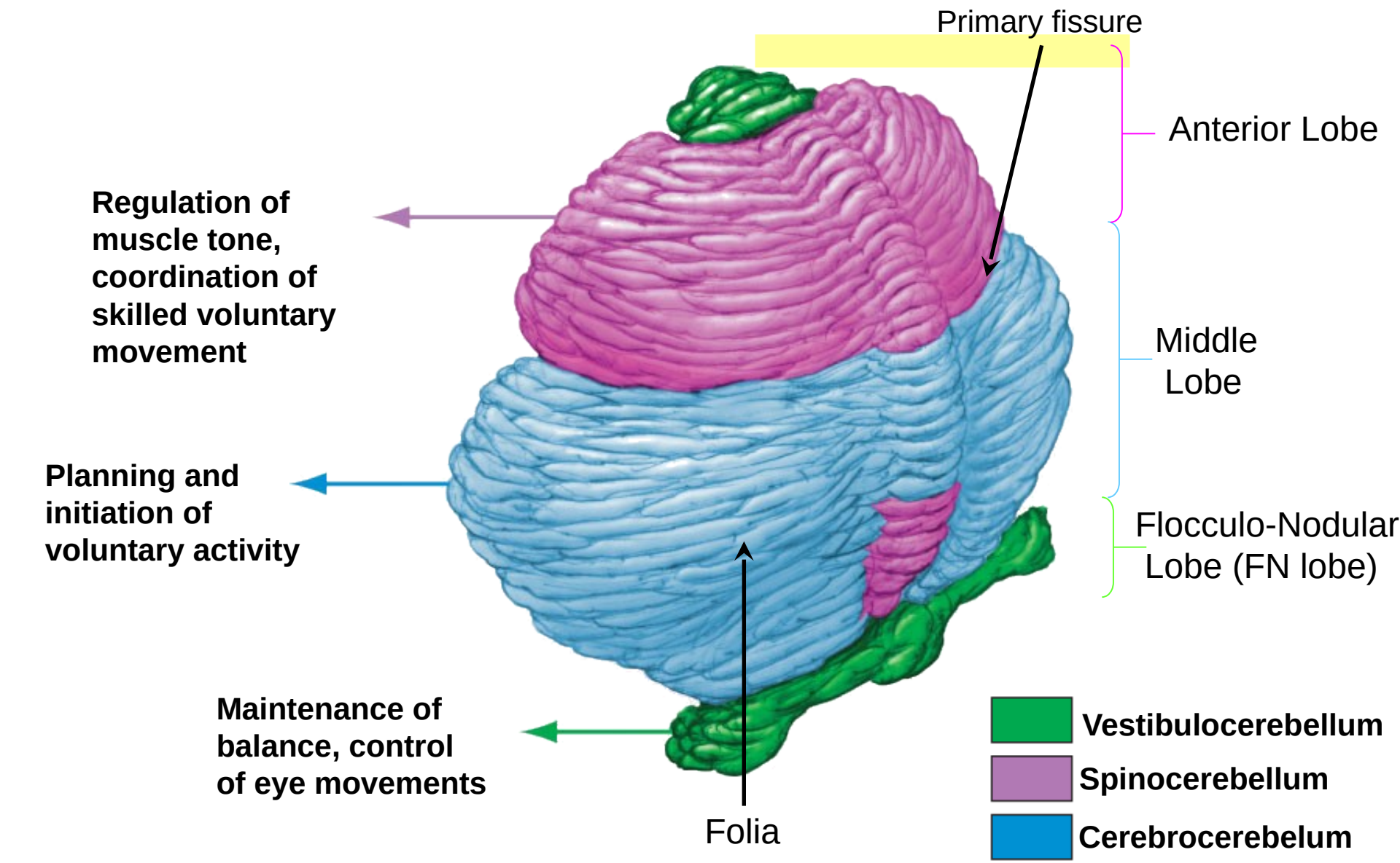
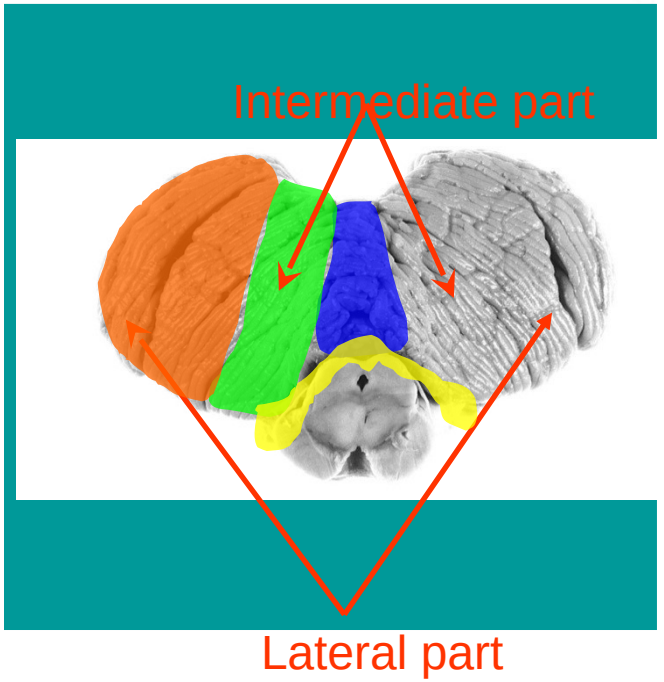

Primary fissure
Anterior Lobe
Middle Lobe
Flocculo-Nodular Lobe (FN lobe)
Regulation of muscle tone, coordination of skilled voluntary movement
Planning and initiation of voluntary activity
Maintenance of balance, control of eye movements
Folia
Vestibulocerebellum
Spinocerebellum
Cerebrocerebellum

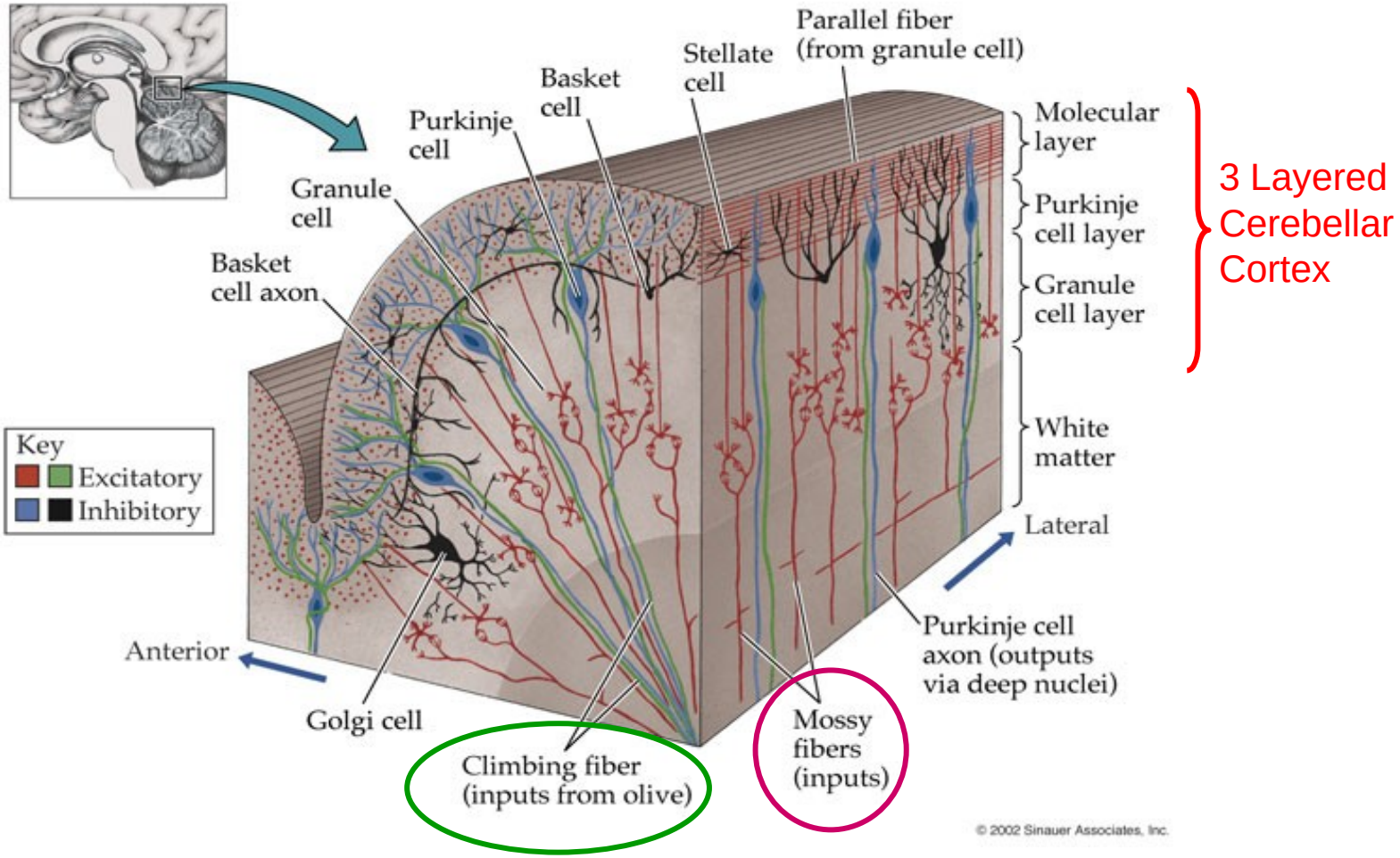
TABLE 15.1 Functional Regions of the Cerebellum

REGION	FUNCTIONS	MOTOR PATHWAYS INFLUENCED
Lateral hemispheres	Motor planning for extremities	Lateral corticospinal tract
Intermediate hemispheres	Distal limb coordination	Lateral corticospinal tract, rubrospinal tract
Vermis and flocculonodular lobe	Proximal limb and trunk coordination	Anterior corticospinal tract, reticulospinal tract, vestibulospinal tract, tectospinal tract
	Balance and vestibulo-ocular reflexes	Medial longitudinal fasciculus



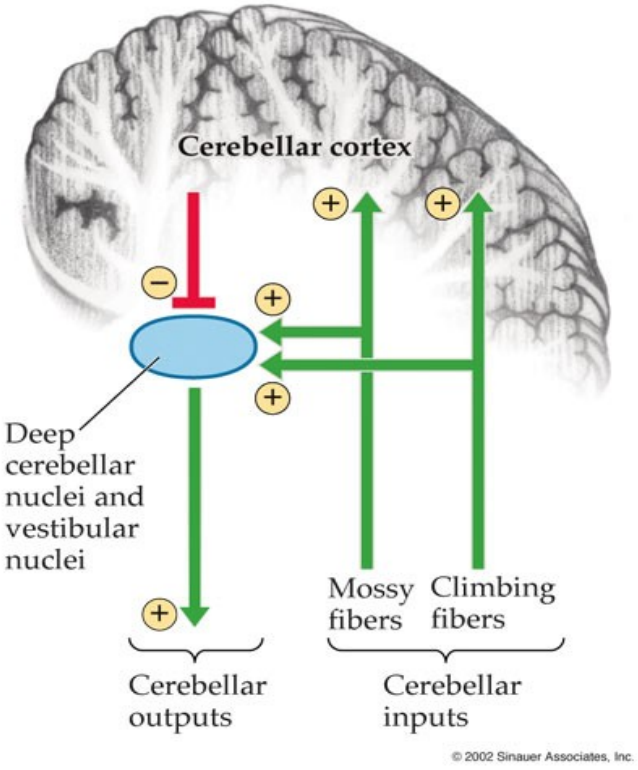
Cerebellum: the Structure

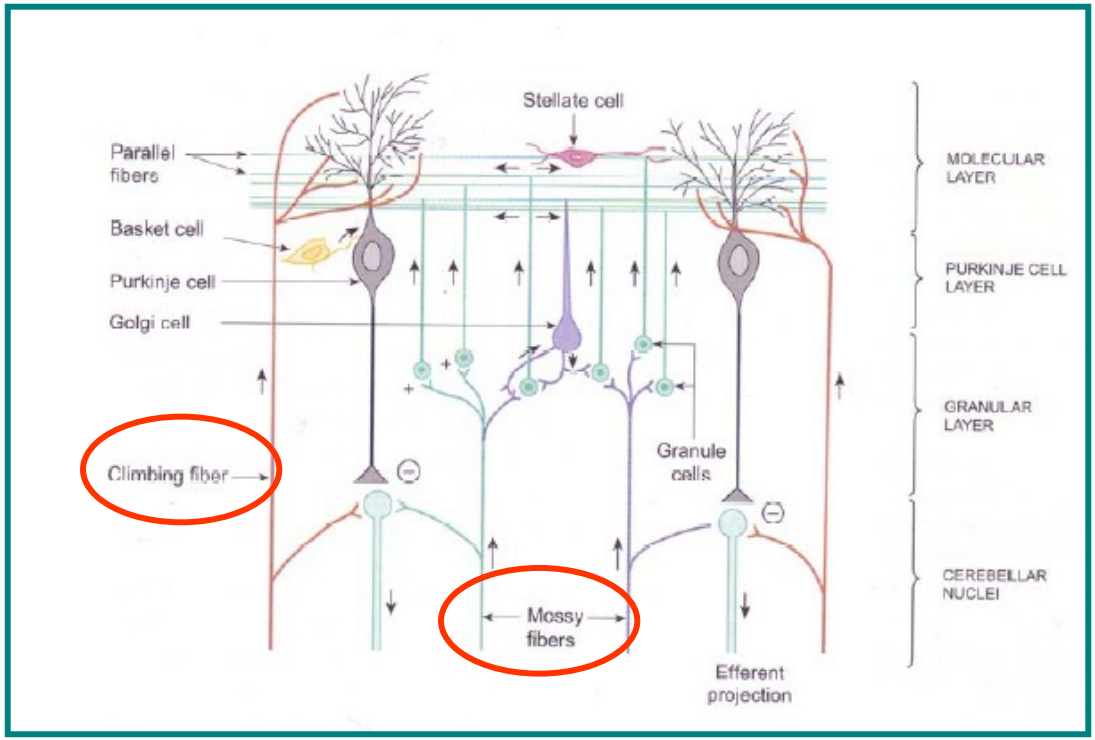
Inputs to the cerebellar cortex: Climbing fibers & Mossy fibers
Climbing fibers: originate in the inferior olive of the medulla
Mossy fibers: originate in all the cerebellar afferent tracts apart from inferior olive
Purkinje cells: The **final** output of the cerebellar cortex



Cerebellum: 3 layered cortex

Climbing fibers: excite the Purkinje cells
Mossy fibers: excite the granule cells
Granule cells: make excitatory contact with the Purkinje cells
Purkinje cells: Tonic inhibition on the activity of the neurons of the cerebellar nuclei
=> All excitatory inputs will be converted to the inhibition
=> Removing the excitatory influence of the cerebellar inputs (erasing)

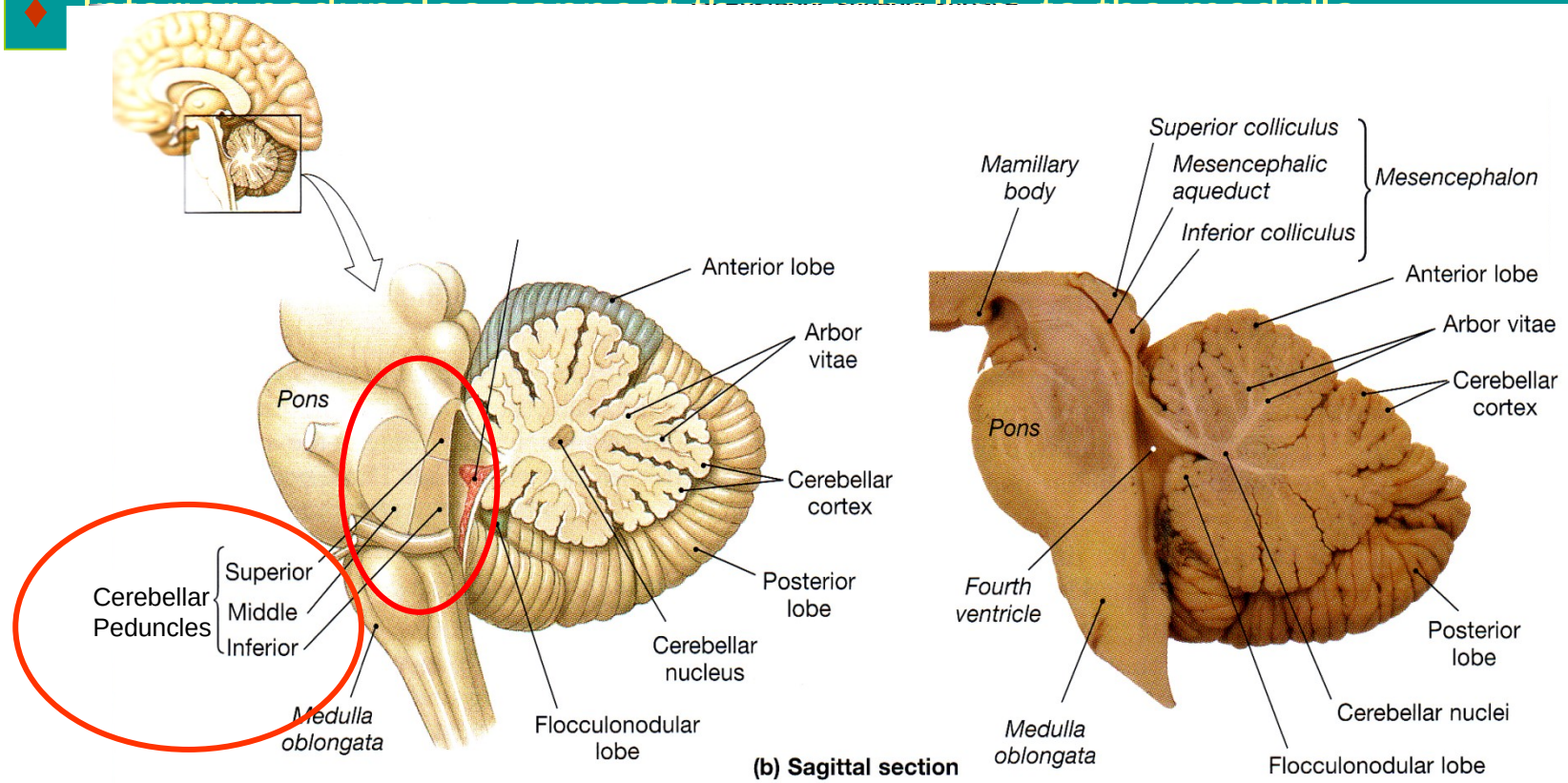




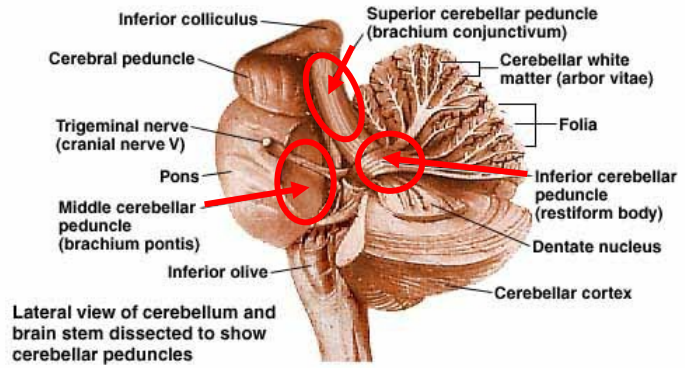
Cerebellar Peduncles

Three paired fiber tracts connect the cerebellum to the brainstem:

- Superior peduncles connect the cerebellum to the midbrain;
- Middle peduncles connect the cerebellum to the pons and to the axis of the brainstem;
- [illegible]



Cerebellar Peduncles



Superior peduncles (to the midbrain):

Fibers originate from neurons in the deep cerebellar nuclei & communicates with the *motor cortex* via the midbrain and the diencephalon (thalamus)

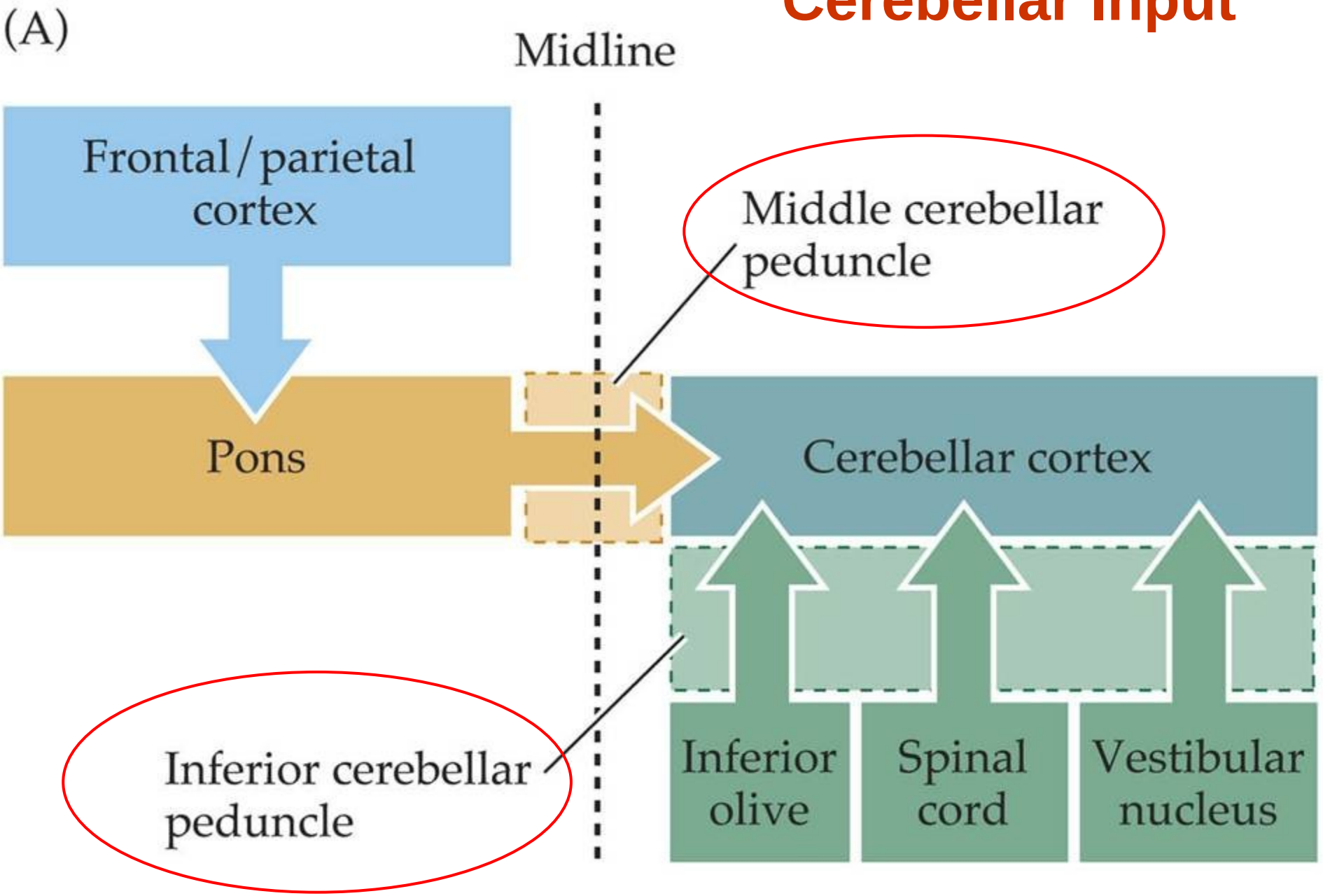
Middle peduncles (to the pons):

Cerebellum receives information advising it of voluntary motor activities initiated by motor cortex

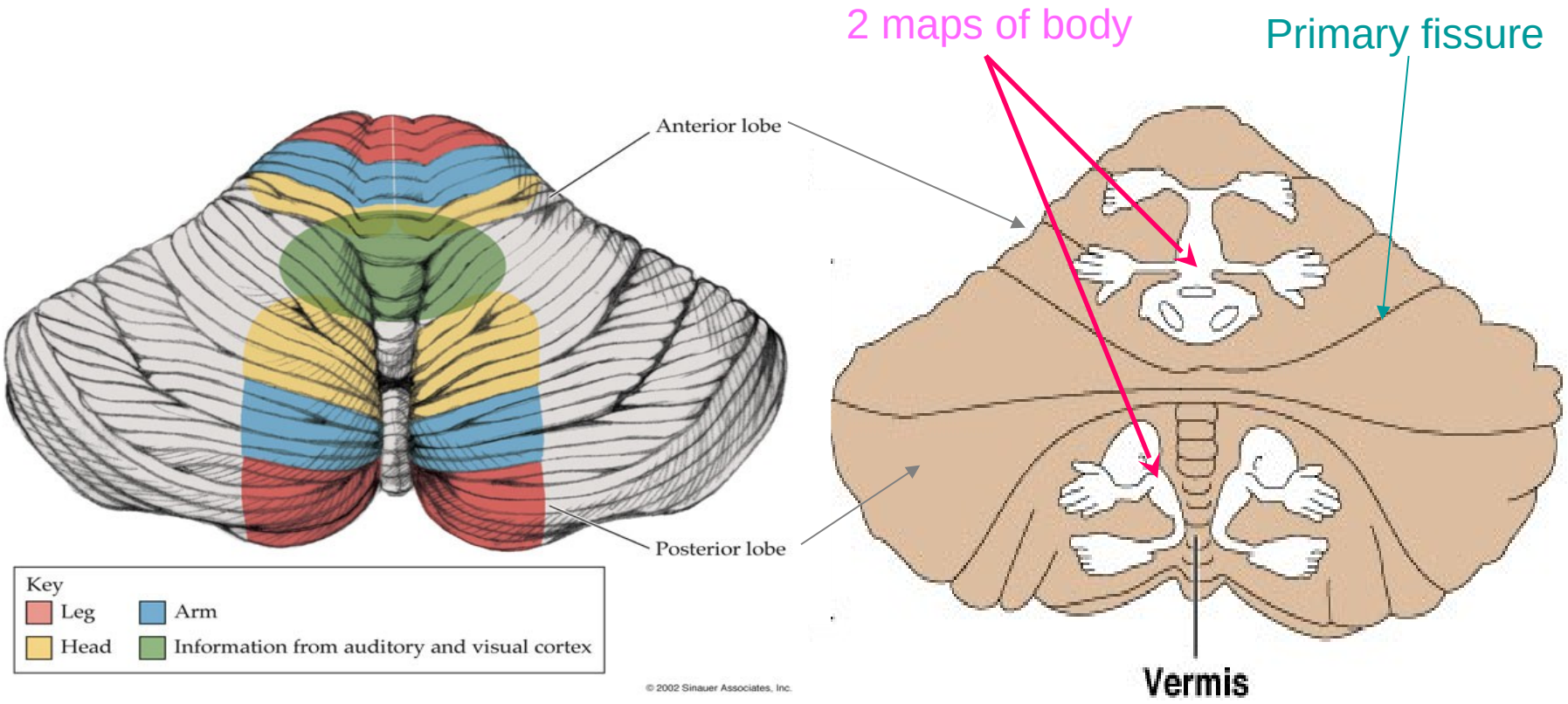
Inferior peduncles (to the medulla):

Afferents conveying sensory information from muscle proprioceptors throughout the body & from the vestibular nuclei of the brainstem (Spinal cord)

Cerebellar Input



Inputs to cerebellum from spinocerebellar tracts have a somatotopic organization.



Signals from the motor cortex, which is also arranged somatotopically, project to corresponding points in the sensory maps of the cerebellum.

Outputs of the Cerebellum

Cerebellar nuclei: dentate, globose, emboliform, fastigial

<u>Dentate nuclei:</u> project contralaterally through the superior cerebellar peduncle to neurons in the contralateral thalamus & from thalamus to motor cortex

Func.: influence planning and initiation of voluntary movement

<u>Emboliform & Globose nuclei:</u> project mainly to the contralateral red nuclei & a small group is projected to the motor cortex

Red Nuclei → Rubrospinal Tract

control of proximal limb muscles

<u>Fastigial nuclei:</u> project to the vestibular nuclei & to the pontine and medullary reticular formation

Vestibulospinal & Reticulospinal tracts

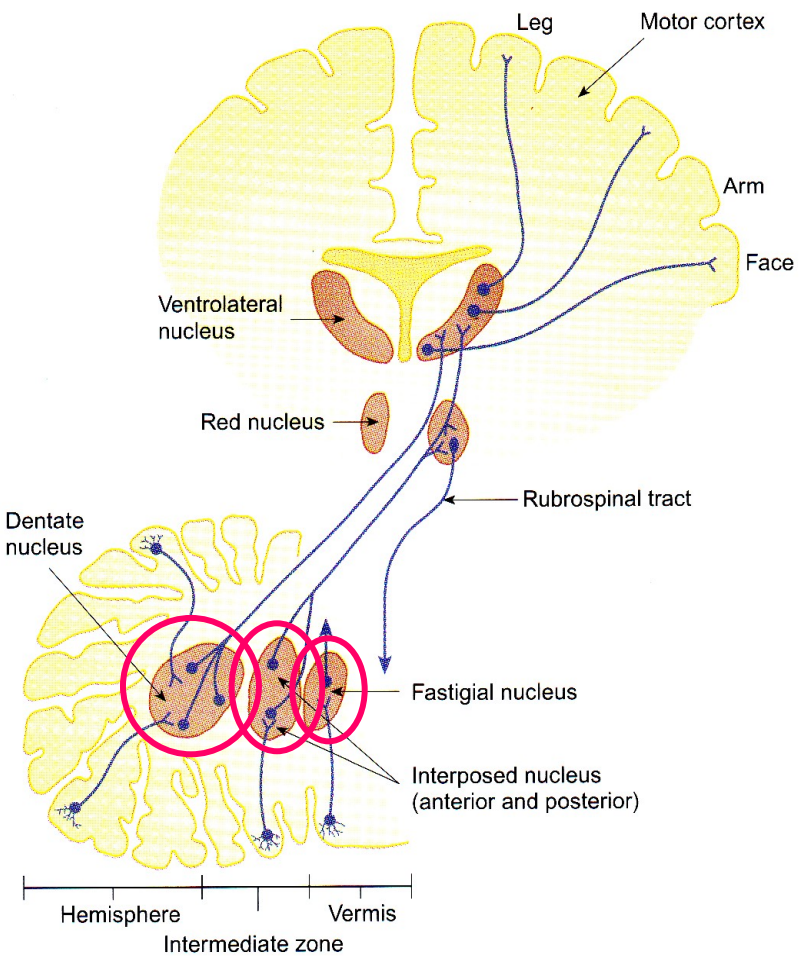


TABLE 15.2 Main Cerebellar Output Pathways

REGION	DEEP NUCLEI	CEREBELLAR PEDUNCLE	MAIN OUTPUT TARGETS OR EQUIVALENT
Lateral hemispheres	Dentate nucleus	Superior cerebellar peduncle	Ventrolateral nucleus of thalamus (VL), parvocellular red nucleus
Intermediate hemispheres	Interposed nuclei	Superior cerebellar peduncle	VL, magnocellular red nucleus
Vermis	Fastigial nuclei	Uncinate fasciculus[a]	VL, tectum
		Juxtarestiform body[b]	Reticular formation, vestibular nuclei
Inferior vermis and flocculonodular lobe	Vestibular nuclei and fastigial nuclei	Juxtarestiform body[b]	Medial longitudinal fasciculus (eye movement pathways)

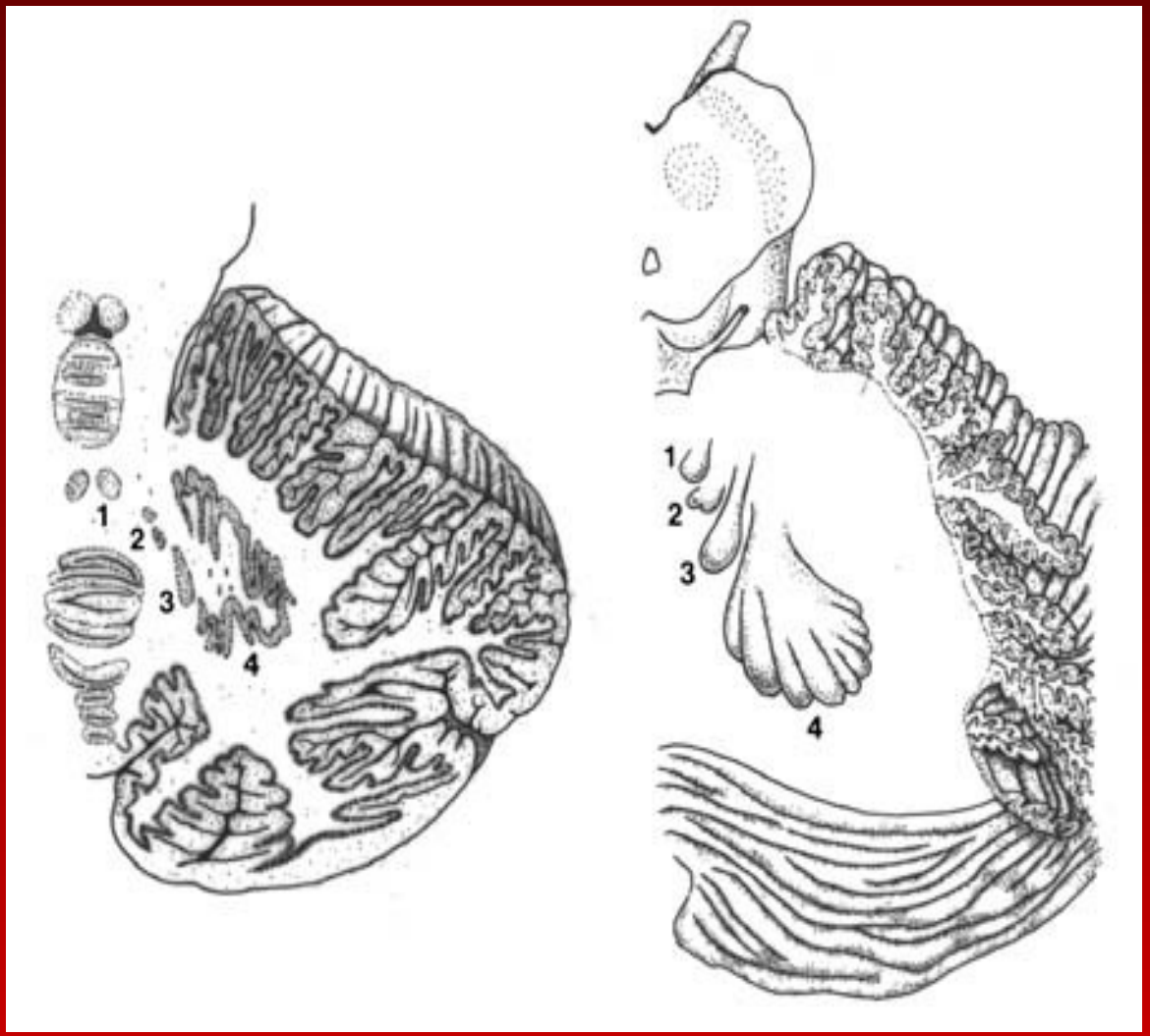
[a]The uncinate fasiculus travels with the superior cerebellar peduncle.
[b]The juxtarestiform body travels with the inferior cerebellar peduncle.

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Deep Cerebellar Nuclei

Fastigial Nuclei

- *Nucleus Interpositus*
  - *Emboliform Nucleus*
  - *Globose Nucleus*
- *Dentate Nucleus*



Deep Nuclei

1. fastigial nucleus
2. globose nucleus
3. emboliform nucleus
4. dentate nucleus

Deep cerebellar nuclei

Sagital section

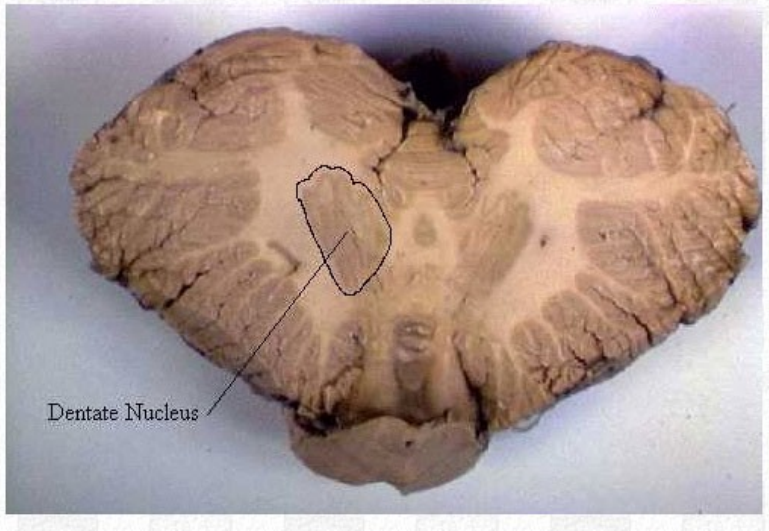


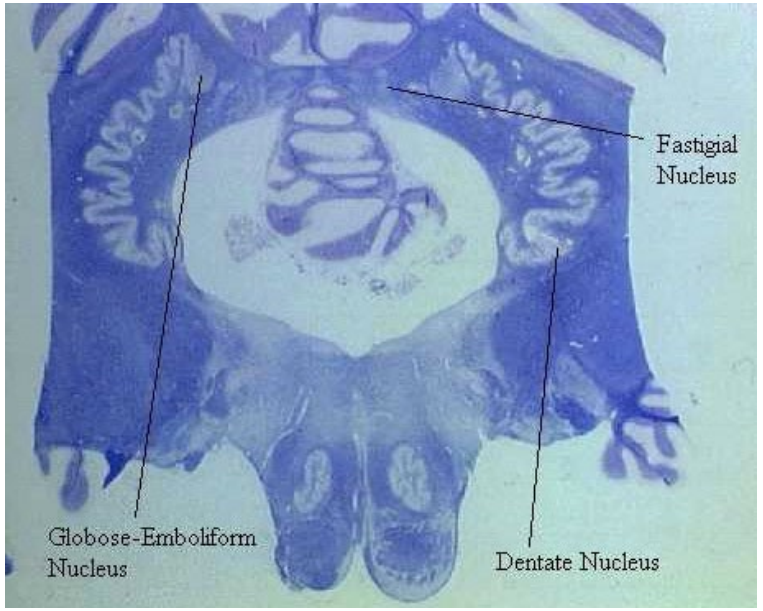
Dentate nucleus

Deep cerebellar nuclei:
Dentate
Interpositus (composed of Globose and Emboliform nuclei)
Fastigial

The deep cerebellar nuclei contain the neurons that project from the cerebellum to other brain regions.
Exception: The neurons of the flocculonodular lobe project directly to the vestibular nuclei and receive afferents from the vestibular labyrinth.

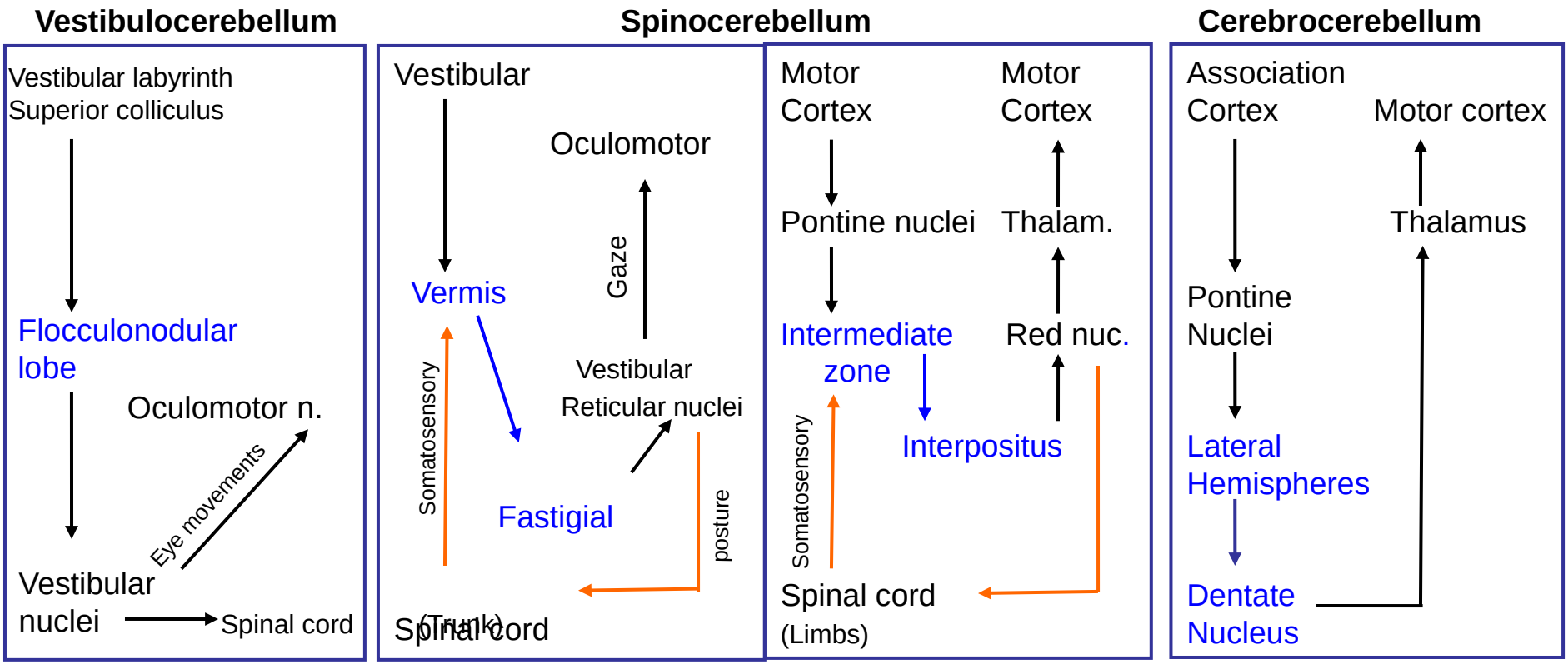
Coronal sections



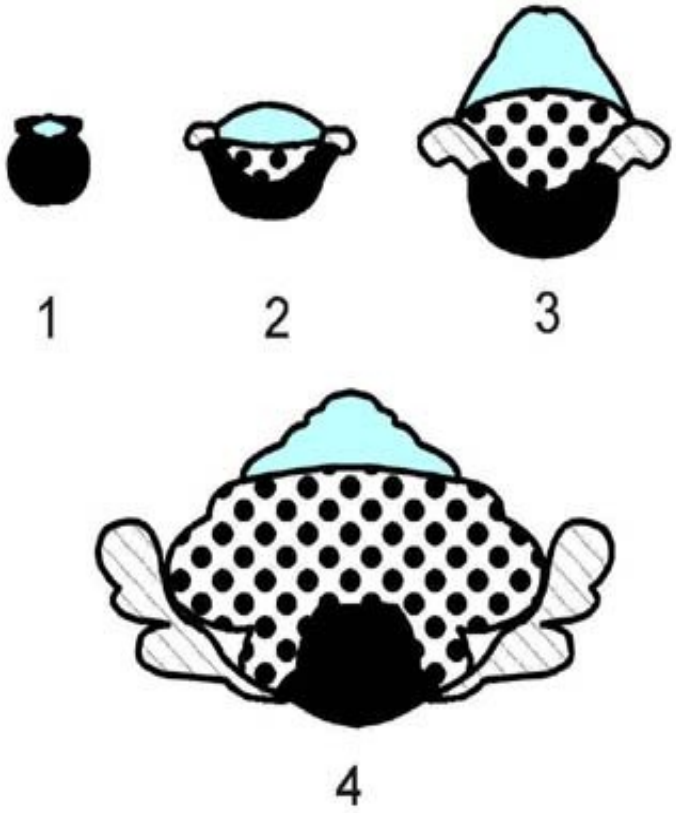


Functional divisions of the cerebellum

1. The vestibulocerebellum comprises the flocculonodular lobe projecting to vestibular nuclei. The flocculonodular lobe receives primary vestibular afferents and projects to the vestibular nuclei. It controls balance and eye movements.
2. Spinocerebellum has two components. I. The vermis which projects to the fastigial nucleus, II. the intermediate zone which projects to the interpositus. It controls posture, locomotion, gaze. Compares the commands emanating from motor cortex with the actual velocity and position of the moving part and corrects signals.
3. Cerebrocerebellum comprises the lateral hemispheres which project to the dentate nucleus. Participates in planning and programming of voluntary, learned, skillful movements that become increasingly precise and rapid with practice.



Evolutionary Subdivision of Cerebellum



Archicerebellum (nodulus)

Archicerebellum (flocculus)

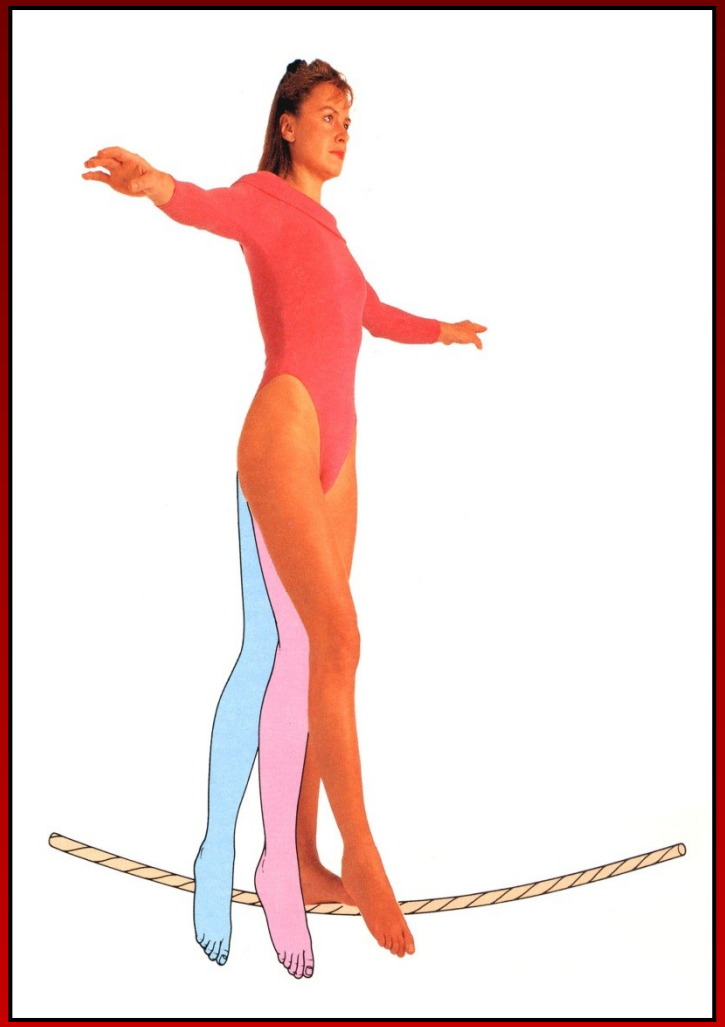
Paleocerebellum

Neocerebellum

Cerebellum ---------- Functions

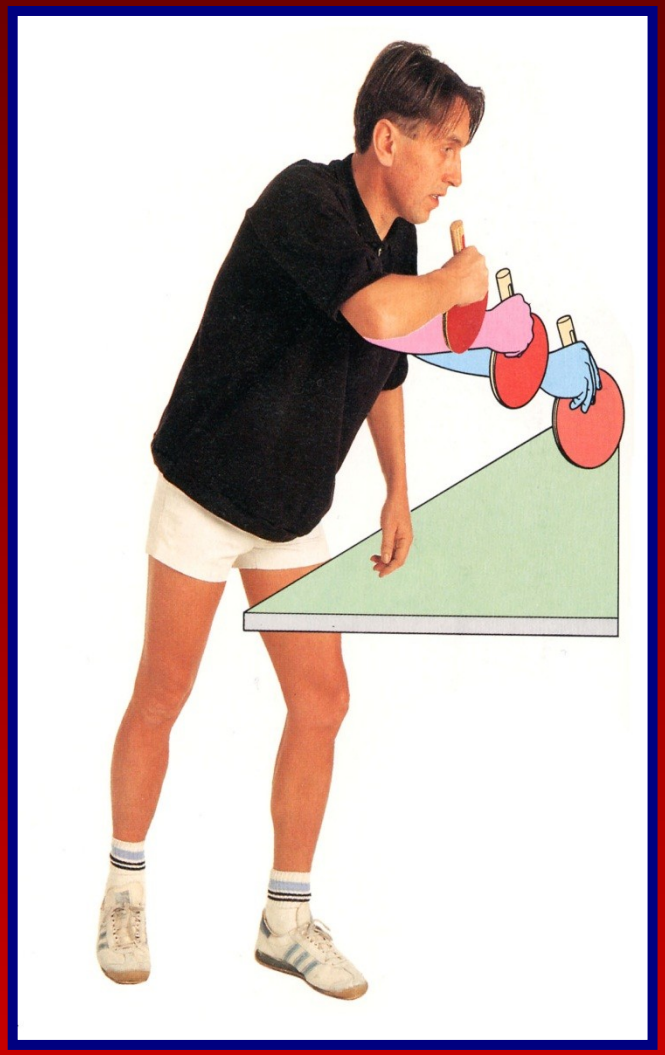
- ***Maintenance of Equilibrium***
  - balance, posture, eye movement
- ***Coordination of half-automatic movement of walking and posture maintenace***
  - posture, gait
- ***Adjustment of Muscle Tone***
- ***Motor Leaning – Motor Skills***
- ***Cognitive Function***

Balance





Motor Skill





Damage to the cerebellum

- Causes a lack of co-ordination:
  - Speaking
  - Walking
  - Tremor

Cerebellar Stroke

- Dizziness, vomiting
- Unsteady so that walking is impossible
- Power, tone and reflexes normal

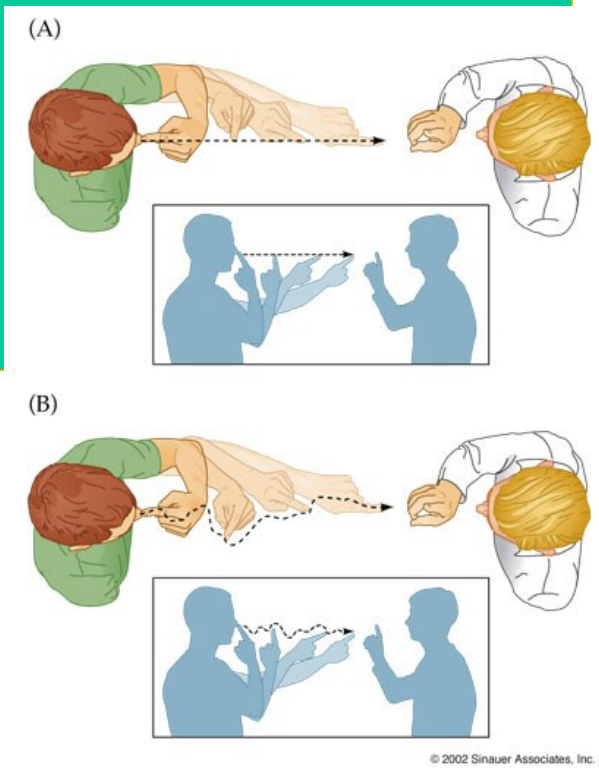
- Area of blood in the cerebellum would show on a CT scan

Clinical Findings and Localization of Cerebellar Lesions

Ataxia refers to disordered contractions of agonist and antagonist muscles and lack of coordination between movements at different joints typically seen in patients with cerebellar lesions.

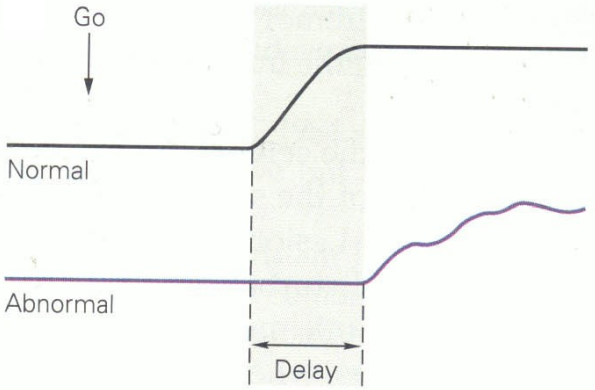
Normal movements require coordination of agonist and antagonist muscles at different joints in order for movement to have smooth trajectory.

In ataxia movements have irregular, wavering course consisting of continuous overshooting, overcorrecting and then overshooting again around the intended trajectory.

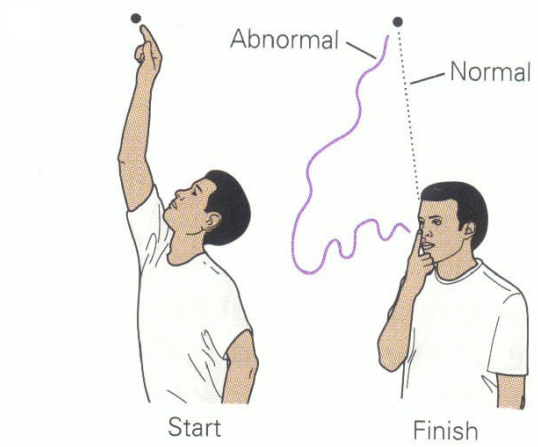


Dysmetria = abnormal undershoot or overshoot during movements toward a target (finger-nose-finger test).

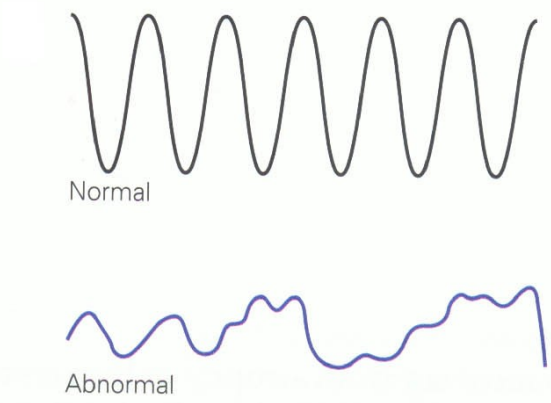
Cerebellar Signs



Hypometria & Response delays



Ataxia



Incoordination/ rapid alternating movements (disdiadocho-kinesia)

Cerebellum

Clinical Syndromes

Ataxia: incoordination of movement

- decomposition of movement

- dysmetria, past-pointing

- dysdiadochokinesia

- rebound phenomenon of Holmes

- gait ataxia, truncal ataxia, titubation

Intention Tremor

Hypotonia, Nystagmus

Archicerebellar Lesion: medulloblastoma

Paleocerebellar Lesion: gait disturbance

Neocerebellar Lesion: hypotonia, ataxia, tremor

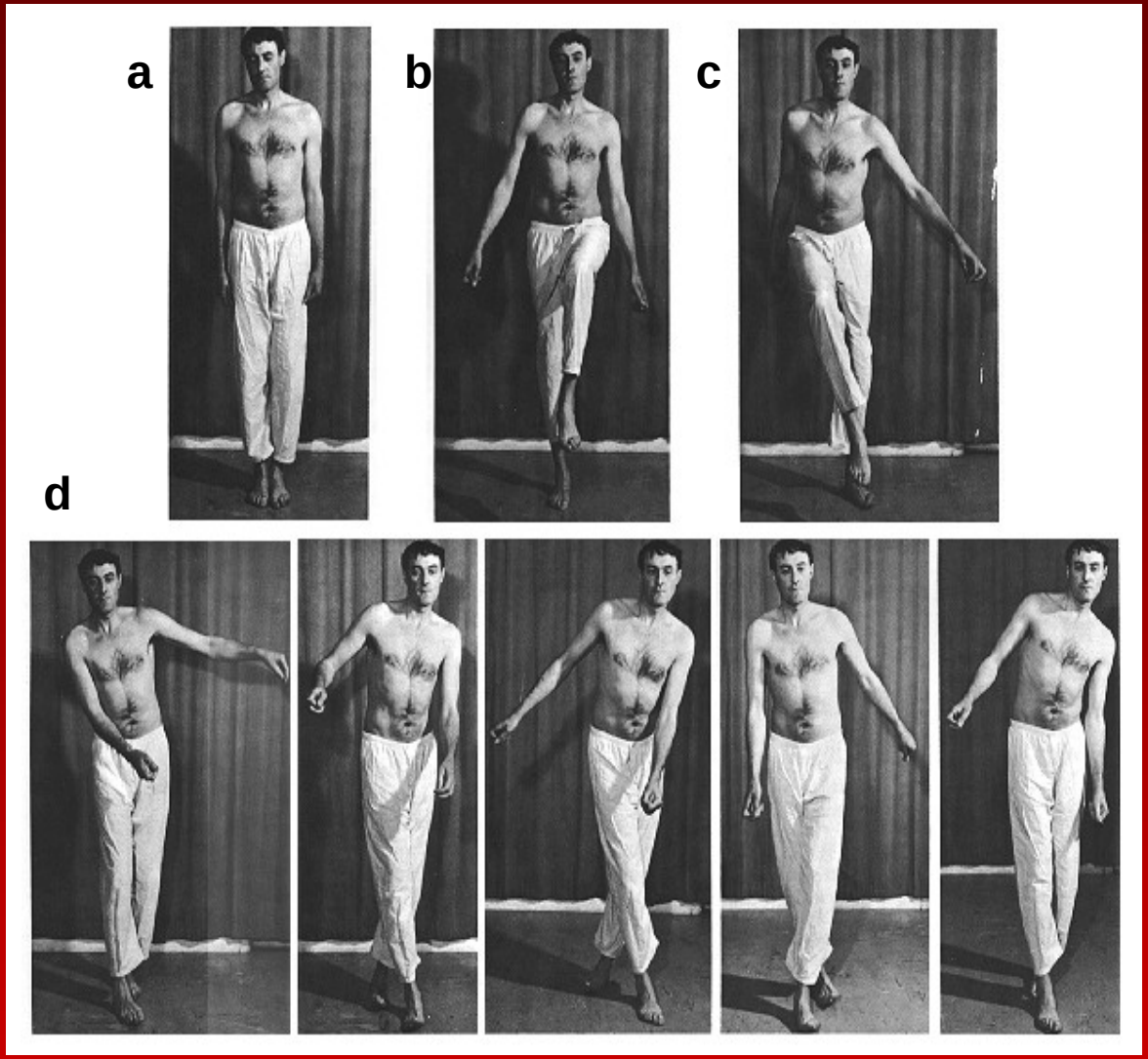




Posture
Gait – Ataxia
Tremor



Cerebellar Ataxia



Ataxic gait and position:
Left cerebellar tumor

a. Sways to the right in standing position

b. Steady on the right leg

c. Unsteady on the left leg

d. ataxic gait

Cerebellar Medulloblastoma

*Cerebellar tumors on **vermis***

- Truncal Ataxia
- Frequent Falling

The child in this picture:

- would not try to stand unsupported
- would not let go of the bed rail if she was stood on the floor.


Cerebellar medulloblastoma

Any Question?

